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Iterated Integrals

1.1 Products of Measure Spaces

Definition 1.1.1 ► product of two σ -algebras; $\mathcal{S} \otimes \mathcal{T}$

Suppose $(X, \mathcal{S})$ and $(Y, \mathcal{T})$ are measurable spaces. Then

- the **product** $\mathcal{S} \otimes \mathcal{T}$ is defined to be the smallest σ -algebra on $X \times Y$ that contains

$$\{A \times B : A \in \mathcal{S}, B \in \mathcal{T}\};$$

- a **measurable rectangle** in $\mathcal{S} \otimes \mathcal{T}$ is a set of the form $A \times B$, where $A \in \mathcal{S}$ and $B \in \mathcal{T}$.

Definition 1.1.2 ► cross sections of sets; $[E]_a$ and $[E]^b$

Suppose X and Y are sets and $E \subset X \times Y$. Then for $a \in X$ and $b \in Y$, the cross sections $[E]_a$ and $[E]^b$ are defined by

$$[E]_a = \{y \in Y : (a, y) \in E\} \quad \text{and} \quad [E]^b = \{x \in X : (x, b) \in E\}.$$

Proposition 1.1.3 ► cross sections of measurable sets are measurable

Suppose $\mathcal{S}$ is a σ -algebra on X and $\mathcal{T}$ is a σ -algebra on Y . If $E \in \mathcal{S} \otimes \mathcal{T}$, then

$$[E]_a \in \mathcal{T} \text{ for every } a \in X \quad \text{and} \quad [E]^b \in \mathcal{S} \text{ for every } b \in Y.$$

Proof sketch.

将截面可测的集合的全体记作 $\mathcal{E}$. 则 $\mathcal{E}$ 是一个 σ -algebra. 可测方块的截面都是可测的, 故 $\mathcal{E}$ 包含了 $\mathcal{S} \otimes \mathcal{T}$.

Definition 1.1.4 ► cross sections of functions; $[f]_a$ and $[f]^b$

Suppose X and Y are sets and $f : X \times Y \rightarrow \mathbf{R}$ is a function. Then for $a \in X$ and $b \in Y$, the cross section functions $[f]_a : Y \rightarrow \mathbf{R}$ and

$[f]^b : X \rightarrow \mathbb{R}$ are defined by

$$[f]_a(y) = f(a, y) \text{ for } y \in Y \quad \text{and} \quad [f]^b(x) = f(x, b) \text{ for } x \in X.$$

Proposition 1.1.5

Suppose $\mathcal{S}$ is a σ -algebra on X and $\mathcal{T}$ is a σ -algebra on Y . Suppose $f : X \times Y \rightarrow \mathbb{R}$ is an $\mathcal{S} \otimes \mathcal{T}$ -measurable function. Then

$[f]_a$ is a $\mathcal{T}$ -measurable function on Y for every $a \in X$

and

$[f]^b$ is an $\mathcal{S}$ -measurable function on X for every $b \in Y$.

Proof sketch.

截面函数的原像集等于原像集的截面:

$$([f]_a)^{-1}(D) = [f^{-1}(D)]_a$$

1.2 Monotone Class Theorem

Definition 1.2.1 ► algebra

Suppose W is a set and $\mathcal{A}$ is a set of subsets of W . Then $\mathcal{A}$ is called an **algebra** on W if the following three conditions are satisfied:

- $\emptyset \in \mathcal{A}$;
- if $E \in \mathcal{A}$, then $W \setminus E \in \mathcal{A}$;
- if E and F are elements of $\mathcal{A}$, then $E \cup F \in \mathcal{A}$.

Proposition 1.2.2 ► the set of finite unions of measurable rectangles is an algebra

Suppose $(X, \mathcal{S})$ and $(Y, \mathcal{T})$ are measurable spaces. Then

- (a) the set of finite unions of measurable rectangles in $\mathcal{S} \otimes \mathcal{T}$ is an algebra on $X \times Y$;
- (b) every finite union of measurable rectangles in $\mathcal{S} \otimes \mathcal{T}$ can be written as a finite union of disjoint measurable rectangles in $\mathcal{S} \otimes \mathcal{T}$.

Definition 1.2.3 ▶ monotone class

Suppose W is a set and $\mathcal{M}$ is a set of subsets of W . Then $\mathcal{M}$ is called a **monotone class** on W if the following two conditions are satisfied:

- If $E_1 \subset E_2 \subset \cdots$ is an increasing sequence of sets in $\mathcal{M}$, then

$$\bigcup_{k=1}^{\infty} E_k \in \mathcal{M};$$

- If $E_1 \supset E_2 \supset \cdots$ is a decreasing sequence of sets in $\mathcal{M}$, then

$$\bigcap_{k=1}^{\infty} E_k \in \mathcal{M}.$$

Theorem 1.2.4 ▶ Monotone Class Theorem

Suppose $\mathcal{A}$ is an algebra on a set W . Then the smallest σ -algebra containing $\mathcal{A}$ is the smallest monotone class containing $\mathcal{A}$.

Proof sketch.

- 设 $\mathcal{M}$ 是包含了 $\mathcal{A}$ 的最小的单调类, $\mathcal{M}$ 包含于 $\sigma(\mathcal{A})$ 是容易的. 主要说明另一边.
- 取 $A \in \mathcal{A}$, 令 $\mathcal{E}$ 是 $\mathcal{M}$ 中在“并 A ”操作下封闭的集合的全体. 则 $\mathcal{E}$ 可以继承 $\mathcal{M}$ 的单调类属性. 从而 $\mathcal{M} \subseteq \mathcal{E}$, $A \cup E \in \mathcal{M}$ 对于所有的 E 成立.
- 令 $\mathcal{D}$ 是 $\mathcal{M}$ 中在“与 $\mathcal{M}$ 中任意元素做并”操作下封闭的集合全体, 则 $\mathcal{D}$ 可以继承 $\mathcal{M}$ 的单调类属性. 则上面的结果就是在说 $A \in \mathcal{D}$, 从而 $\mathcal{A} \subseteq \mathcal{D}$. 得到 $\mathcal{M} \subseteq \mathcal{D}$. 这就是说 $\mathcal{M}$ 在有限并下封闭.
- 有限并加上对递增列封闭, 就是对可数并封闭.
- 令 $\mathcal{M}'$ 是 $\mathcal{M}$ 中在“补集”操作下封闭的全体, 则 $\mathcal{A} \subseteq \mathcal{M}'$. $\mathcal{M}'$ 可以继承 $\mathcal{M}$ 的单调类属性 (可以认为单调类的定义是关于补集对偶的). 故 $\mathcal{M} \subseteq \mathcal{M}'$, $\mathcal{M}$ 在补集下封闭.

1.3 Products of Measures

Theorem 1.3.1 ▶ measure of cross section is a measurable function

Suppose $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ -finite measure spaces. If $E \in \mathcal{S} \otimes \mathcal{T}$, then

1. $x \mapsto \nu([E]_x)$ is an $\mathcal{S}$ -measurable function on X ;
2. $y \mapsto \mu([E]^y)$ is a $\mathcal{T}$ -measurable function on Y .

Proof sketch.

考虑 (a)

先考虑有限测度的情况下.

- 令 $\mathcal{M}$ 是具有性质 (a) 的 E 的全体.
- 所有可测方块具有性质 (a).
- 令 $\mathcal{A}$ 表示可测方块生成的 algebra. $[\cdot]_x$ 是保持集合运算的, 可以由此将 $[E]_x$ 按无交并的可测方块的并分解 $[E_1]_x \cup [E_2]_x \cup \cdots \cup [E_n]_x$.
- 由此, $E \in \mathcal{A}$ 的函数 $x \mapsto \nu([E]_x)$ 是 $x \mapsto \nu([E_i]_x)$ 的和, 是可测的. 故 $\mathcal{A} \subseteq \mathcal{M}$.
- ν 的自上自下连续性, 可测函数对逐点极限的封闭性 (需要测度是有限的), 给出 $\mathcal{M}$ 是单调类.
- 由单调类定理, $\mathcal{M}$ 包含 $\mathcal{S} \otimes \mathcal{T}$.

推广到 σ -有限的情况, 只需要取出 Y 的一系列递增的有限测度分解 $Y_1 \subseteq Y_2 \subseteq \cdots$, $\bigcup_{k=1}^{\infty} Y_k = Y$. 做逼近 $\nu([E]_x) = \lim_{k \rightarrow \infty} \nu([E \cap (X \times Y_k)]_x)$ 即可.

Definition 1.3.2 ▶ product of two measures; $\mu \times \nu$

Suppose $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ -finite measure spaces. For $E \in \mathcal{S} \otimes \mathcal{T}$, define $(\mu \times \nu)(E)$ by

$$(\mu \times \nu)(E) = \int_X \int_Y \chi_E(x, y) d\nu(y) d\mu(x).$$

Theorem 1.3.3 ▶ product of two measures is a measure

Suppose $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ -finite measure spaces. Then $\mu \times \nu$ is a measure on $(X \times Y, \mathcal{S} \otimes \mathcal{T})$.

Proof sketch.

利用单调收敛定理即可说明可数可加性.

1.4 Iterated Integrals

Theorem 1.4.1 ▶ Tonelli's Theorem

Suppose $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ -finite measure spaces. Suppose $f : X \times Y \rightarrow [0, \infty]$ is $\mathcal{S} \otimes \mathcal{T}$ -measurable. Then

(a) $x \mapsto \int_Y f(x, y) d\nu(y)$ is an $\mathcal{S}$ -measurable function on X ,

(b) $y \mapsto \int_X f(x, y) d\mu(x)$ is a $\mathcal{T}$ -measurable function on Y ,

and

$$\int_{X \times Y} f d(\mu \times \nu) = \int_X \int_Y f(x, y) d\nu(y) d\mu(x) = \int_Y \int_X f(x, y) d\mu(x) d\nu(y).$$

Proof sketch.

先说明 $f = \chi_E$ 的情况.

- 这个情况下 (a) (b) 就是 Theorem 1.3.1. 令 $\mathcal{M}$ 是使得 χ_E 满足积分交换性的集合.
- 可测方块是属于 $\mathcal{M}$ 的, 进而可测方块生成的代数 $\mathcal{A}$ 含于 $\mathcal{M}$.
- 先考虑测度 μ, ν 有限的情况. 由单调收敛定理, $\mathcal{M}$ 对递增集合列封闭. 当 μ, ν 测度有限时, 由控制收敛定理 $\mathcal{M}$ 对递减列封闭. 故此时 $\mathcal{M}$ 是单调类.
- 由单调类定理, 当 μ, ν 有限时, $\mathcal{M}$ 包含 $\mathcal{S} \otimes \mathcal{T}$.
- 推广到 σ -有限的情况, 做分解 $X_1 \subseteq X_2 \cdots \subseteq X, Y_1 \subseteq Y_2 \cdots \subseteq Y$. 则对于 $E \in \mathcal{S} \otimes \mathcal{T}$, $(E \cap (X_j \times Y_k)) \in \mathcal{M}$. 单调收敛定理可以给出 $E \cap (X \times Y_k) \in \mathcal{M}$, 再用一次单调收敛定理给出 $E = E \cap (X \times Y) \in \mathcal{M}$.

对于 f 是一般的非负可测函数的情况, 找一系列趋于 f 的递增的简单函数 $\{f_k\}$. 题述性质对 f_k 是成立的. 单调收敛定理给出 (a) 和 (b). 再用单调收敛定理穿过两层积分 $\int_X \int_Y$ 给出积分的交换性.

Theorem 1.4.2 ▶ Fubini's Theorem

Suppose $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ -finite measure spaces. Suppose

$$f : X \times Y \rightarrow [-\infty, \infty]$$

is $\mathcal{S} \otimes \mathcal{T}$ -measurable and

$$\int_{X \times Y} |f| d(\mu \times \nu) < \infty.$$

Then

$$\int_Y |f(x, y)| d\nu(y) < \infty \text{ for almost every } x \in X$$

and

$$\int_X |f(x, y)| d\mu(x) < \infty \text{ for almost every } y \in Y.$$

Furthermore,

(a) $x \mapsto \int_Y f(x, y) d\nu(y)$ is an $\mathcal{S}$ -measurable function on X ,

(b) $y \mapsto \int_X f(x, y) d\mu(x)$ is a $\mathcal{T}$ -measurable function on Y ,

and

$$\int_{X \times Y} f d(\mu \times \nu) = \int_X \int_Y f(x, y) d\nu(y) d\mu(x) = \int_Y \int_X f(x, y) d\mu(x) d\nu(y).$$

Proof sketch.

- 对 $|f|$ 使用 Tonelli 定理中的积分等式, 得到累次绝对积分的可积性, 同时得到单边绝对积分的几乎处处有限.
- 对 f^+, f^- 使用 Tonelli 定理中的积分函数的可测性结论, 并由可积性可知未定义的部分是零测集, 得到 $x \mapsto \int_Y f(x, y) d\nu(y)$ 的可测性.
- 将 f 做正负部分解, 对正负部积分分别使用 Tonelli 定理得到积分等式.

1.5 Area Under the Graph

Definition 1.5.1 ▶ region under the graph

Suppose X is a set and $f : X \rightarrow [0, \infty]$ is a function. Then the **region under the graph** of f , denoted U_f , is defined by

$$U_f = \{(x, t) \in X \times (0, \infty) : 0 < t < f(x)\}.$$

Proposition 1.5.2 ▶ area under the graph of a function equals the integral

Suppose $(X, \mathcal{S}, \mu)$ is a σ -finite measure space and $f : X \rightarrow [0, \infty]$ is an $\mathcal{S}$ -measurable function. Let $\mathcal{B}$ denote the σ -algebra of Borel subsets of $(0, \infty)$, and let λ denote Lebesgue measure on $((0, \infty), \mathcal{B})$. Then $U_f \in \mathcal{S} \otimes \mathcal{B}$ and

$$(\mu \times \lambda)(U_f) = \int_X f d\mu = \int_{(0, \infty)} \mu(\{x \in X : t < f(x)\}) d\lambda(t).$$

Proof sketch.

令 $F(x, t) = f(x) - t$, 则 U_f 是 $F^{-1}((0, \infty))$ 是可测的. 积分等式由 Tonelli 定理得到.